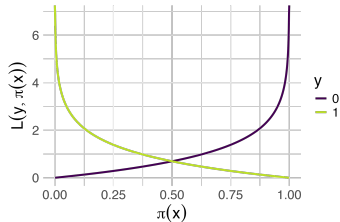


Introduction to Machine Learning

Advanced Risk Minimization

Logistic regression (Deep-Dive)



Learning goals

- Derive the gradient of the logistic regression
- Derive the Hessian of the logistic regression
- Show that the logistic regression is a convex problem

LOGISTIC REGRESSION: RISK PROBLEM

Given n observations $(\mathbf{x}^{(i)}, y^{(i)}) \in \mathcal{X} \times \mathcal{Y}$ with $\mathcal{X} = \mathbb{R}^d, \mathcal{Y} = \{0, 1\}$ we want to minimize the risk

$$\mathcal{R}_{\text{emp}} = - \sum_{i=1}^n y^{(i)} \log(\pi(\mathbf{x}^{(i)} | \boldsymbol{\theta})) + (1 - y^{(i)}) \log(1 - \pi(\mathbf{x}^{(i)} | \boldsymbol{\theta}))$$

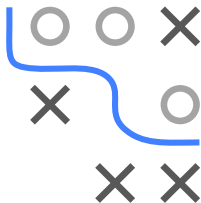
with respect to $\boldsymbol{\theta}$ where the probabilistic classifier

$$\pi(\mathbf{x}^{(i)} | \boldsymbol{\theta}) = s(f(\mathbf{x}^{(i)} | \boldsymbol{\theta}))$$

the sigmoid function $s(f) = \frac{1}{1 + \exp(-f)}$ and the score $f(\mathbf{x}^{(i)} | \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$

NB: Note that $\frac{\partial}{\partial f} s(f) = s(f)(1 - s(f))$ and $\frac{\partial f(\mathbf{x}^{(i)} | \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = (\mathbf{x}^{(i)})^\top$

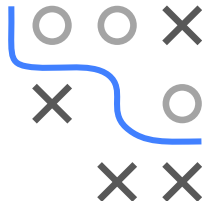
From now on we abbreviate $\pi(\mathbf{x}^{(i)} | \boldsymbol{\theta})$ as $\pi_{\boldsymbol{\theta}}^{(i)}$



LOGISTIC REGRESSION: GRADIENT

We find the gradient of logistic regression with the chain rule:

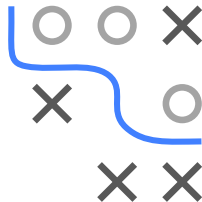
$$\begin{aligned}\frac{\partial}{\partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}} &= - \sum_{i=1}^n \frac{\partial}{\partial \pi_{\boldsymbol{\theta}}^{(i)}} y^{(i)} \log(\pi_{\boldsymbol{\theta}}^{(i)}) \frac{\partial \pi_{\boldsymbol{\theta}}^{(i)}}{\partial \boldsymbol{\theta}} + \\ &\quad \frac{\partial}{\partial \pi_{\boldsymbol{\theta}}^{(i)}} (1 - y^{(i)}) \log(1 - \pi_{\boldsymbol{\theta}}^{(i)}) \frac{\partial \pi_{\boldsymbol{\theta}}^{(i)}}{\partial \boldsymbol{\theta}} \\ &= - \sum_{i=1}^n \frac{y^{(i)}}{\pi_{\boldsymbol{\theta}}^{(i)}} \frac{\partial \pi_{\boldsymbol{\theta}}^{(i)}}{\partial \boldsymbol{\theta}} - \frac{1 - y^{(i)}}{1 - \pi_{\boldsymbol{\theta}}^{(i)}} \frac{\partial \pi_{\boldsymbol{\theta}}^{(i)}}{\partial \boldsymbol{\theta}} \\ &= - \sum_{i=1}^n \left(\frac{y^{(i)}}{\pi_{\boldsymbol{\theta}}^{(i)}} - \frac{1 - y^{(i)}}{1 - \pi_{\boldsymbol{\theta}}^{(i)}} \right) \frac{\partial s(f(\mathbf{x}^{(i)} | \boldsymbol{\theta}))}{\partial f(\mathbf{x}^{(i)} | \boldsymbol{\theta})} \frac{\partial f(\mathbf{x}^{(i)} | \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \\ &= - \sum_{i=1}^n (y^{(i)}(1 - \pi_{\boldsymbol{\theta}}^{(i)}) - (1 - y^{(i)})\pi_{\boldsymbol{\theta}}^{(i)}) (\mathbf{x}^{(i)})^\top\end{aligned}$$



LOGISTIC REGRESSION: GRADIENT

$$= \sum_{i=1}^n (\pi_{\theta}^{(i)} - y^{(i)}) (\mathbf{x}^{(i)})^{\top}$$

$$= (\pi(\mathbf{X} | \theta) - \mathbf{y})^{\top} \mathbf{X}$$



where

- $\mathbf{X} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)})^\top \in \mathbb{R}^{n \times d}$
- $\mathbf{y} = (y^{(1)}, \dots, y^{(n)})^\top$
- $\pi(\mathbf{X} | \theta) = (\pi_\theta^{(i)}[1], \dots, \pi_\theta^{(i)}[n])^\top \in \mathbb{R}^n$

$\implies$ The gradient $\nabla_{\theta} \mathcal{R}_{\text{emp}} = \left(\frac{\partial}{\partial \theta} \mathcal{R}_{\text{emp}} \right)^{\top} = \mathbf{X}^{\top} (\pi(\mathbf{X} | \theta) - \mathbf{y})$

This formula can now be used in gradient descent and its friends

LOGISTIC REGRESSION: HESSIAN

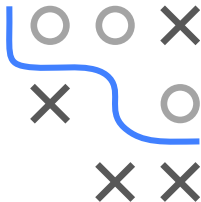
We find the Hessian via differentiation:

$$\begin{aligned}\nabla_{\boldsymbol{\theta}}^2 \mathcal{R}_{\text{emp}} &= \frac{\partial^2}{\partial \boldsymbol{\theta}^\top \partial \boldsymbol{\theta}} \mathcal{R}_{\text{emp}} = \frac{\partial}{\partial \boldsymbol{\theta}^\top} \sum_{i=1}^n (\pi_{\boldsymbol{\theta}}^{(i)} - y^{(i)}) (\mathbf{x}^{(i)})^\top \\ &= \sum_{i=1}^n \mathbf{x}^{(i)} (\pi_{\boldsymbol{\theta}}^{(i)} (1 - \pi_{\boldsymbol{\theta}}^{(i)})) (\mathbf{x}^{(i)})^\top \\ &= \mathbf{X}^\top \mathbf{D} \mathbf{X}\end{aligned}$$

where $\mathbf{D} \in \mathbb{R}^{n \times n}$ is a diagonal matrix with diagonal

$$\left(\pi_{\boldsymbol{\theta}}^{(1)} [1] (1 - \pi_{\boldsymbol{\theta}}^{(1)} [1]), \dots, \pi_{\boldsymbol{\theta}}^{(n)} [n] (1 - \pi_{\boldsymbol{\theta}}^{(n)} [n]) \right)$$

Can now be used in Newton-Raphson and other 2nd order optimizers



LOGISTIC REGRESSION: CONVEXITY

Finally, we check that logistic regression is a convex problem:

We define the diagonal matrix $\bar{\mathbf{D}} \in \mathbb{R}^{n \times n}$ with diagonal

$$\left(\sqrt{\pi_{\theta}^{(i)}[1](1 - \pi_{\theta}^{(i)}[1])}, \dots, \sqrt{\pi_{\theta}^{(i)}[n](1 - \pi_{\theta}^{(i)}[n])} \right)$$

which is possible since π maps into $(0, 1)$

With this, we get for any $\mathbf{w} \in \mathbb{R}^d$ that

$$\mathbf{w}^\top \nabla_{\theta}^2 \mathcal{R}_{\text{emp}} \mathbf{w} = \mathbf{w}^\top \mathbf{X}^\top \bar{\mathbf{D}}^\top \bar{\mathbf{D}} \mathbf{X} \mathbf{w} = (\bar{\mathbf{D}} \mathbf{X} \mathbf{w})^\top \bar{\mathbf{D}} \mathbf{X} \mathbf{w} = \|\bar{\mathbf{D}} \mathbf{X} \mathbf{w}\|_2^2 \geq 0$$

since obviously $\mathbf{D} = \bar{\mathbf{D}}^\top \bar{\mathbf{D}}$

$\implies \nabla_{\theta}^2 \mathcal{R}_{\text{emp}}$ is positive semi-definite $\implies \mathcal{R}_{\text{emp}}$ is convex

